

ABDULHAKIM ABDURRAZAK SULAIMAN

Python Backend Engineer | Distributed Systems & AI Integration

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 **GitHub:** <https://github.com/Abdoul2146> |  **LinkedIn:** <https://www.linkedin.com/in/abdulhakim-abdurrazak-sulaiman-98bb38242/>

PROFESSIONAL SUMMARY

Backend Software Engineer specializing in Python and distributed backend systems with hands-on experience building scalable APIs, asynchronous task processing, workflow automation, and AI-powered applications. Experienced designing multi-tenant architectures using FastAPI, SQLAlchemy, PostgreSQL, Celery, Redis, and AWS S3 while integrating LLMs including OpenAI, Gemini, and Whisper into production systems. Strong advocate of Test-Driven Development (TDD), clean architecture, and high-performance backend design, with experience delivering secure, reliable systems from architecture through deployment.

TECHNICAL SKILLS

- Languages: Python (expert), JavaScript/TypeScript, Dart SQL, Bash
- Backend: FastAPI, Flask, SQLAlchemy, Alembic, Celery, Redis, WebSocket, JWT, REST APIs, OAuth2
- Frontend & Mobile: React Native, Flutter, HTML, CSS
- Testing: OpenAPI/Swagger, Unittest, Pytest
- Databases and Storage: PostgreSQL, SQLite, MongoDB, Redis, Amazon S3, Cloudinary
- AI/LLM: TensorFlow, scikit-learn, Whisper, Gemini API, PySpark, Pandas, NumPy.
- Hosting Services/tools: Render, Railway, Linux
- Others: Git, REST API design, system architecture

EDUCATION

Ahmadu Bello University, Zaria

- B.Eng in Computer Engineering
2025. CGPA(4.01/5.00)

EXPERIENCE

- **Backend Developer: Cuoral Technologies (Remote) | Dec 2025 – Present**
 - Engineered multi-channel communication integrations (X/Twitter, Telegram, Facebook/Instagram) with OAuth 2.0 PKCE flows and encrypted credential management, enabling customers to manage support across 5+ social platforms from a unified dashboard.
 - Designed and implemented ticket routing system with custom tag management, allowing organizations to auto-route incoming tickets by customer attributes and custom metadata.
 - Built comprehensive survey implementation system for post-interaction feedback collection and analysis, integrated with workflow pipelines.
 - Architected a configurable workflow automation engine supporting conditional branching, delayed execution, reusable workflow templates, execution monitoring, and real-time analytics.
 - Developed robust test suite (pytest/unittest) covering 1500+ test cases across integration and unit tests for APIs, CRUD operations, and business logic validation.
 - Designed and implemented asynchronous background task processing using **Celery** and **Redis** for workflow automation, customer engagement campaigns, scheduled survey execution, and long-running business processes.
 - Leveraged **Redis** for distributed caching and low-latency application state management, including in-app survey trigger evaluation and workflow execution
 - Integrated **Amazon S3** object storage for secure media/file management across customer communication workflows.
- **ALX Software Engineering (Backend Specialization) Jan 2023 – Feb 2025**
 - Completed 12-month intensive software engineering program with 3-months specialization backend track: Python, SQL, CI/CD.
 - Delivered Several Python/backend based Major/mini-projects
- **SIWES Industrial Training – Full-Stack Developer May 2024 – Nov 2024**
 - NetConnect Server (FastAPI + WebSocket): real-time LAN messaging for 200+ users.
 - Implemented JWT auth, group chat, and file transfer APIs.
 - Designed SQLite Schemas.

PROJECTS

- **English to Hausa SPEECH (Full-Stack ML/AI Engineer, FastAPI, NextJS)**
 - Architected production-grade FastAPI backend with three independent, fully-decoupled processing pipelines: video-to-audio extraction, English-to-Hausa machine translation, and speech synthesis (TTS).
 - Engineered asynchronous job processing system with PostgreSQL persistence, enabling long-running GPU workloads (2+ hours) with real-time progress tracking, status polling, and result retrieval via RESTful APIs.
 - Integrated AI/ML models (OpenAI Whisper for ASR, Google Gemini for translation, Meta MMS-TTS for synthesis) with lazy-loading singleton pattern to optimize GPU memory management and model caching.
 - Implemented secure API authentication via custom X-API-KEY headers to prevent resource exhaustion on GPU-constrained environments (Lightning AI GPU Studio).
 - Designed decoupled architecture with clean domain layering (models, schemas, CRUD, services, endpoints) enabling independent operation or chaining of pipeline stages; fully recoverable from restarts with no in-memory dependencies.
 - Developed comprehensive job management system with automatic workspace cleanup, file lifecycle management, and optional cloud storage integration (Cloudinary) for seamless media delivery.
 - Deployed to Lightning AI GPU Studio with 40GB persistent storage, managing model downloads, checkpoint handling, and FFmpeg/yt-dlp integration for multi-platform video support.

GitHub: https://github.com/Abdoul2146/english_to_hausa_backend & https://github.com/Abdoul2146/english_to_hausa_frontend

- **NetConnect:- LAN Messaging and file sharing(FastAPI + WebSocket)**
 - Modular REST/WebSocket API for offline LAN chat with user auth, group rooms, file transfer (chunked).
 - SQLite schema: users, rooms, messages, files.
- **CCMT Crop Disease Dashboard API (PySpark, Scikit-learn, MongoDB Atlas, Streamlit)**
 - Designed and implemented an end-to-end machine learning pipeline using PySpark for distributed data preprocessing, cleaning, and feature engineering on large Kaggle crop disease datasets.
 - Trained and evaluated a Random Forest model using Scikit-learn for crop disease prediction and exposed real-time inference through a RESTful API.
 - Stored model evaluation metrics in MongoDB Atlas and developed an interactive Streamlit dashboard for prediction visualization and analytics.

GitHub: <https://github.com/Abdoul2146/CCMT-Crop-Disease-Pest--Big-Data-Project>

- **PaddleOCR PDF Text Extraction App (Flask)**
 - Optical Character Recognition Web App for extracting texts from Pdfs and images
 - Integrated paddle's kits/sdk for the OCR tasks
- **Encryption/Decryption Web App (Flask)**
 - Made use of Flask for the development
 - Used Cryptography modules for encryption and decryption processes

GitHub: https://github.com/Elder2146/File_Encryption

- **Humor Detection Service (Python + TensorFlow)**
 - Trained and test 3-5 ML models (BiGRU, CNN-BiLSTM, ColBERT) with 86% accuracy.

GitHub: https://github.com/Abdoul2146/Humor_Detection_Techniques

CERTIFICATIONS

- Software Engineering & AI:
 - ALX Software Engineering Program (Backend) – ALX Africa | Jan 2025
 - AI Career Essentials – ALX Africa | Nov 2024
- Huawei Certified ICT Associate (HCIA) – Huawei Technologies
 - Linux Distribution (openEuler) – Nov 2024
 - Relational Database (openGauss) – Dec 2024
- Cybersecurity:
 - Google Cybersecurity Certificate – Coursera | Oct 2024

SOFT SKILLS:

- Team Collaboration - Research-Oriented - Problem Solving - Technical Communication